SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:		**Reg. No.:**	
Section / Batch:		**Date:**	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
	GRAND TOTAL:	**100 Marks**	

SECTION A – APPLICATION BASED QUESTIONS

Q1. Online Examination Evaluation System An online examination system evaluates student responses using the recurrence $T(n) = 3T(n/3) + n^2$. Apply the Master Theorem to determine the time complexity. Identify parameters and justifythe case selection. Analyze how performance changes as the number of answer scripts increases.Interpret efficiency using asymptotic notation.

Online Examination Evaluation System

Recurrence: $T(n) = 3T(n/3) + n^2$

1. Understanding of Problem Context

- The online examination system evaluates n answer scripts by recursively dividing the workload into 3 equal subproblems, each of size n/3.

- After the subproblems are solved, an additional cost of n^2 is required to combine/aggregate the evaluated results (e.g., score compilation, cross-checking).
- This models a divide-and-conquer evaluation strategy, similar to how large exam datasets are split across evaluators and merged back.

2. Application of Concepts

Step 1: Identify Master Theorem parameters

- $a = 3$ → number of subproblems per recursive call
- $b = 3$ → factor by which problem size shrinks
- $f(n) = n^2$ → cost of work done outside recursion (combining step)

Step 2: Compute $n^{(\log_b a)}$

- $\log_b a = \log_3 3 = 1$
- $n^{(\log_b a)} = n^1 = n$

Step 3: Compare $f(n)$ with $n^{(\log_b a)}$

- $f(n) = n^2$ vs $n^1 = n$
- n^2 grows strictly faster than n, so $f(n) = \Omega(n^{(1+\varepsilon)})$ with $\varepsilon = 1$

3. Problem-Solving Approach

- Since $f(n)$ dominates $n^{(\log_b a)}$, this falls under Case 3 of the Master Theorem.
- Regularity condition must be verified for Case 3 to apply:
- $a \cdot f(n/b) \leq c \cdot f(n)$ for some constant $c < 1$
- $3 \cdot (n/3)^2 = 3 \cdot (n^2/9) = n^2/3$
- $n^2/3 = (1/3) \cdot n^2$ → $c = 1/3$, which is less than 1 ✔ Condition satisfied
- Hence, Case 3 is valid and applicable here.

4. Accuracy of Solution

- Applying Master Theorem Case 3: $T(n) = \Theta(n^2)$
- This means the running time is asymptotically bound tightly by n^2, both from above ($O(n^2)$) and below ($\Omega(n^2)$).

5. Clarity — Performance Analysis & Final Interpretation

As the number of answer scripts (n) increases, the evaluation time grows quadratically, since the n^2 combining step dominates over the recursive splitting.

This shows the system is efficient for moderate-sized datasets but scales poorly for very large-scale examinations.

Final Answer: $T(n) = \Theta(n^2)$ — Big-O: $O(n^2)$, Big-Ω: $\Omega(n^2)$, Big-Θ: $\Theta(n^2)$

Q2. AI Model Training Pipeline An AI training system processes datasets using the recurrence $T(n) = 9T(n/3) + n^2$. Apply the Master Theorem to determine the time complexity. Clearly identify parameters and justify the selected case. Analyze how training time grows with dataset size. Interpret efficiency using asymptotic notation.

AI Model Training Pipeline

Recurrence: $T(n) = 9T(n/3) + n^2$

1. Understanding of Problem Context

- The AI training pipeline processes a dataset of size n by splitting it into 9 subproblems, each of size n/3 (e.g., parallel feature extraction across 9 partitions).
- An additional cost of n^2 is incurred after recursion for tasks like gradient aggregation or model synchronization.
- This reflects a higher branching factor compared to Q1, representing a more computation-heavy training structure.

2. Application of Concepts

Step 1: Identify Master Theorem parameters

- $a = 9 \rightarrow$ number of subproblems
- $b = 3 \rightarrow$ shrink factor
- $f(n) = n^2 \rightarrow$ combining cost

Step 2: Compute n^(log_b a)

- $\log_b a = \log_3 9 = 2$
- $n^{\log_b a} = n^2$

Step 3: Compare f(n) with n^(log_b a)

- $f(n) = n^2$ and $n^{\log_b a} = n^2$
- Both are asymptotically equal: $f(n) = \Theta(n^{\log_b a})$

3. Problem-Solving Approch

- Since f(n) exactly matches n^(log_b a), this falls under Case 2 of the Master Theorem.
- Case 2 formula: $T(n) = \Theta(n^{\log_b a} \cdot \log n)$
- No regularity condition check is needed here (only required for Case 3).

4. Accuracy of Solution

- Applying Master Theorem Case 2: $T(n) = \Theta(n^2 \log n)$
- This is a tight bound — both upper and lower bounds match with an extra logarithmic factor compared to plain n^2.

5. Clarity — Performance Analysis & Final Interpretation

- As dataset size (n) increases, training time grows as $n^2 \log n$, which is slightly worse than pure n^2 growth (seen in Q1).
- The higher branching factor ($a = 9$) increases the total recursive work, adding the log n multiplier even though the combining cost f(n) is the same as Q1.
- Final Answer: $T(n) = \Theta(n^2 \log n)$ — Big-O: $O(n^2 \log n)$, Big-Ω: $\Omega(n^2 \log n)$, Big-Θ: $\Theta(n^2 \log n)$

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________